# Midterm

## HD Sheets

## 2024-09-11

##DSE 5001 Mid term

We will use some data from the fivethirtyeight site, and from some other sources

Prep for the exam by making sure you can do all these problems, and that you understand the content of each data set.

You may want to do some research on your own to understand these data sets a bit, and to be sure you understand the ideas in each problem.

Prepare an RMD with answers to all of these problems. You may use the RMD and other sources of information during the exam. You can look things up during the exam, but you cannot asking questions of another living person, or of LLM models like ChatGPT or Claude.

(If you want to try to ask questions of non-living person's that's your call, but most ghosts just don't know much R, as far as I can tell).

Turn in your prepared RMD with all answers prepared and annotated at the start of the exam. The preparation is 40% of the exam grade.

Then you will get 3 randomly chosen questions from this set of questions to answer on the exam during the one-hour exam window.

The questions during the live exam will be modified, so while you can (and should) cut and paste answers from your preparation RMD, you will need to modify your answers to reflect the changes in the exam question

I might alter the exam question by

a.) changing the data set b.) changing what variables I want to see plotted, or in a table c.) changing ranges of values

I will use the Quiz function in Canvas to randomly give you three problems, cut and paste these into an RMD and then use your prepared code to answer them. You can just paste the altered problems right into your prepared RMD if that helps.

I will answer questions about the problem statements, but will not tell you if you have a problem correct or not.

Most of the questions are derived from homework and/or PairProgramming examples from Modules 1 to 4

Module 5 content is not covered on this exam, the final will be Modules 5 to 7

This exam thus has a "take home preparation" portion and a "live" portion. Due to the sheer volume of material in the class and the rapid pace, this format will reward extensive preparation work, and lower the impact of work during the one hour exam window. At the same time, you do have to be able to execute the code quickly, demonstrating that you understand how it works.

# Exam Time

The exam will open at 5 pm EST Sept 26 and close at 9pm Sept27, you must complete the exam in a one-hour long block within that time frame

Note: I have be available on Zoom from 7 to 9 pm EST on Thursday Sept 26 and 7-9 pm EST on Friday Sept 27, so that you can jump on zoom if you have to ask me questions. I strongly urge you to take the exam during one of the times when I am online. I cannot guarantee you will be able to ask me questions at other times during the exam window. If you feel you will want to be able to ask questions, try to plan on taking the exam between 7-9 pm EST on Thursday or Friday.

```
library("fivethirtyeight")
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tidyr)
library(ggplot2)
```

#Steak Survey, Problem 1

A consumer preference data set

a.) Make sure all the factors are set

```
library(dplyr)
str(steak_survey)
```

```
## tibble [550 × 15] (S3: tbl_df/tbl/data.frame)
##  $ respondent_id: num [1:550] 3.24e+09 3.23e+09 3.23e+09 3.23e+09 3.23e+09 ...
##  $ lottery_a    : logi [1:550] FALSE TRUE TRUE FALSE FALSE TRUE ...
##  $ smoke        : logi [1:550] NA FALSE FALSE TRUE FALSE FALSE ...
##  $ alcohol      : logi [1:550] NA TRUE TRUE TRUE TRUE FALSE ...
##  $ gamble       : logi [1:550] NA FALSE TRUE TRUE FALSE FALSE ...
##  $ skydiving    : logi [1:550] NA FALSE FALSE FALSE FALSE FALSE ...
##  $ speed        : logi [1:550] NA FALSE TRUE TRUE TRUE TRUE ...
##  $ cheated      : logi [1:550] NA FALSE TRUE TRUE TRUE FALSE ...
##  $ steak        : logi [1:550] NA TRUE TRUE TRUE TRUE TRUE ...
##  $ steak_prep   : Ord.factor w/ 5 levels "Rare"<"Medium rare"<..: NA 2 1 3 3 2 NA 2 3
2 ...
##  $ female       : logi [1:550] NA FALSE FALSE FALSE FALSE FALSE ...
##  $ age          : Ord.factor w/ 4 levels "18-29"<"30-44"<..: NA 4 4 4 4 1 4 1 1 4 ...
##  $ hhold_income : Factor w/ 6 levels "$0 - $24,999",..: NA 3 5 3 3 1 5 2 3 2 ...
##  $ educ         : Ord.factor w/ 5 levels "Less than high school degree"<..: NA 3 5 4
5 3 5 3 4 3 ...
##  $ region       : chr [1:550] NA "East North Central" "South Atlantic" "New England"
...
```

```
help("steak_survey")
library(tidyr)
```

b.) How many respondents smoke, drink and gamble?

```
smoke_count <- steak_survey |>
  filter(smoke == "TRUE") |>
  nrow()

alcohol_count <- steak_survey |>
  filter(alcohol == "TRUE") |>
  nrow()

gamble_count <- steak_survey |>
  filter(gamble == "TRUE") |>
  nrow()

cat("smoke:", smoke_count, "\n")
```

```
## smoke: 84
```

```
cat("drink:", alcohol_count, "\n")
```

```
## drink: 416
```

```
cat("gamble:", gamble_count, "\n")
```

```
## gamble: 257
```

c.) Make a table or report that show as a cross table of counts by values in the categories smoke, drink and gamble

```
cross_table <- steak_survey |>
  group_by(smoke, alcohol, gamble) |>
  summarize(count = n(), .groups = 'drop')

print(cross_table)
```

```
## # A tibble: 15 × 4
##    smoke alcohol gamble count
##    <lgl> <lgl>   <lgl>  <int>
##  1 FALSE FALSE   FALSE     80
##  2 FALSE FALSE   TRUE      29
##  3 FALSE FALSE   NA         2
##  4 FALSE TRUE    FALSE    174
##  5 FALSE TRUE    TRUE     167
##  6 FALSE TRUE    NA         1
##  7 TRUE  FALSE   FALSE      8
##  8 TRUE  FALSE   TRUE       3
##  9 TRUE  TRUE    FALSE     16
## 10 TRUE  TRUE    TRUE      57
## 11 NA    FALSE   FALSE      1
## 12 NA    FALSE   TRUE       1
## 13 NA    FALSE   NA         1
## 14 NA    TRUE    FALSE      1
## 15 NA    NA      NA         9
```

d.) How many regions are there? Don’t count this up from a table, show a function to do this.

```
regions <- steak_survey %>%
  summarise(unique_regions = n_distinct(region))

regions
```

```
## # A tibble: 1 × 1
##   unique_regions
##            <int>
## 1             10
```

e.) select all the female respondents who don’t drink, but do gamble and are from the second highest income category

```
steak_survey |>
  filter(female == "TRUE" & alcohol == "FALSE" & gamble == "TRUE" &
  hhold_income == levels(hhold_income)[length(levels(hhold_income)) - 1]
  )
```

```
## # A tibble: 0 × 15
## # ℹ 15 variables: respondent_id <dbl>, lottery_a <lgl>, smoke <lgl>,
## #   alcohol <lgl>, gamble <lgl>, skydiving <lgl>, speed <lgl>, cheated <lgl>,
## #   steak <lgl>, steak_prep <ord>, female <lgl>, age <ord>, hhold_income <fct>,
## #   educ <ord>, region <chr>
```

f.) Select all the people under 45 with a Bachelors degree or higher

```
steak_survey |>
  filter(age < "45-60" & (educ == "Bachelor degree" | educ == "Graduate degree"))
```

```
## # A tibble: 133 × 15
##    respondent_id lottery_a smoke alcohol gamble skydiving speed cheated steak
##            <dbl> <lgl>     <lgl> <lgl>   <lgl>  <lgl>     <lgl> <lgl>   <lgl>
##  1    3234953052 TRUE      TRUE  TRUE    TRUE   FALSE     TRUE  FALSE   TRUE
##  2    3234948197 TRUE      FALSE FALSE   TRUE   FALSE     TRUE  FALSE   TRUE
##  3    3234947477 FALSE     FALSE TRUE    TRUE   TRUE      TRUE  FALSE   TRUE
##  4    3234946739 TRUE      TRUE  TRUE    TRUE   FALSE     TRUE  FALSE   TRUE
##  5    3234941615 TRUE      FALSE TRUE    TRUE   FALSE     TRUE  FALSE   TRUE
##  6    3234939289 TRUE      FALSE TRUE    TRUE   FALSE     FALSE FALSE   TRUE
##  7    3234937720 FALSE     FALSE FALSE   FALSE  FALSE     TRUE  FALSE   TRUE
##  8    3234937709 TRUE      FALSE TRUE    TRUE   FALSE     TRUE  FALSE   TRUE
##  9    3234930461 TRUE      FALSE FALSE   FALSE  FALSE     TRUE  FALSE   TRUE
## 10    3234928903 FALSE     TRUE  TRUE    TRUE   FALSE     TRUE  FALSE   FALSE
## # ℹ 123 more rows
## # ℹ 6 more variables: steak_prep <ord>, female <lgl>, age <ord>,
## #   hhold_income <fct>, educ <ord>, region <chr>
```

# Steak Survey, Problem 2

A consumer preference data set

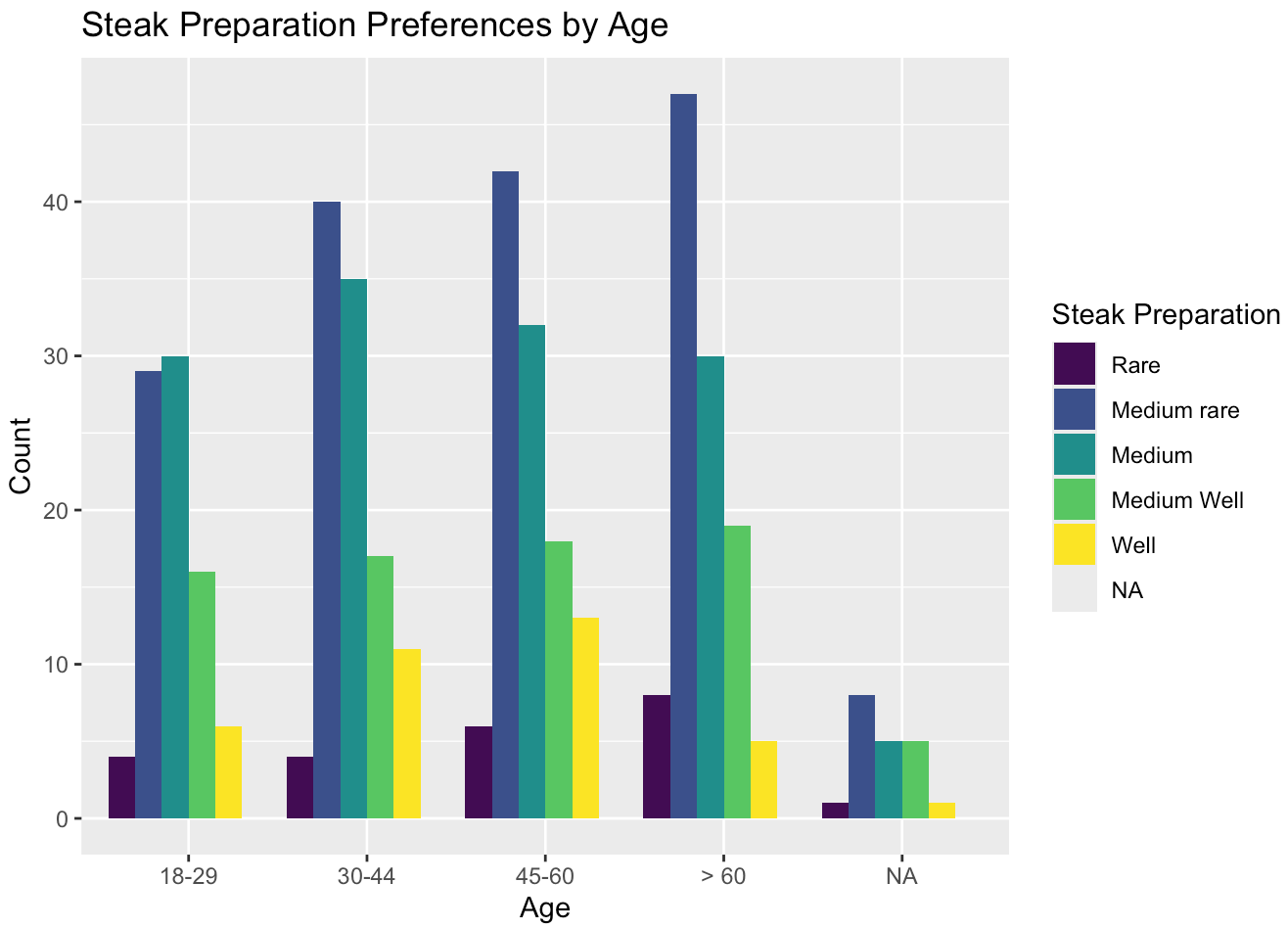
```
head(steak_survey)
```

```
## # A tibble: 6 × 15
##   respondent_id lottery_a smoke alcohol gamble skydiving speed cheated steak
##           <dbl> <lgl>     <lgl> <lgl>   <lgl>  <lgl>     <lgl> <lgl>   <lgl>
## 1    3237565956 FALSE     NA    NA      NA     NA        NA    NA      NA
## 2    3234982343 TRUE      FALSE TRUE    FALSE  FALSE     FALSE FALSE   TRUE
## 3    3234973379 TRUE      FALSE TRUE    TRUE   FALSE     TRUE  TRUE    TRUE
## 4    3234972383 FALSE     TRUE  TRUE    TRUE   FALSE     TRUE  TRUE    TRUE
## 5    3234958833 FALSE     FALSE TRUE    FALSE  FALSE     TRUE  TRUE    TRUE
## 6    3234955240 TRUE      FALSE FALSE   FALSE  FALSE     TRUE  FALSE   TRUE
## # ℹ 6 more variables: steak_prep <ord>, female <lgl>, age <ord>,
## #   hhold_income <fct>, educ <ord>, region <chr>
```

Make plots that shows steak prep preferences depending on

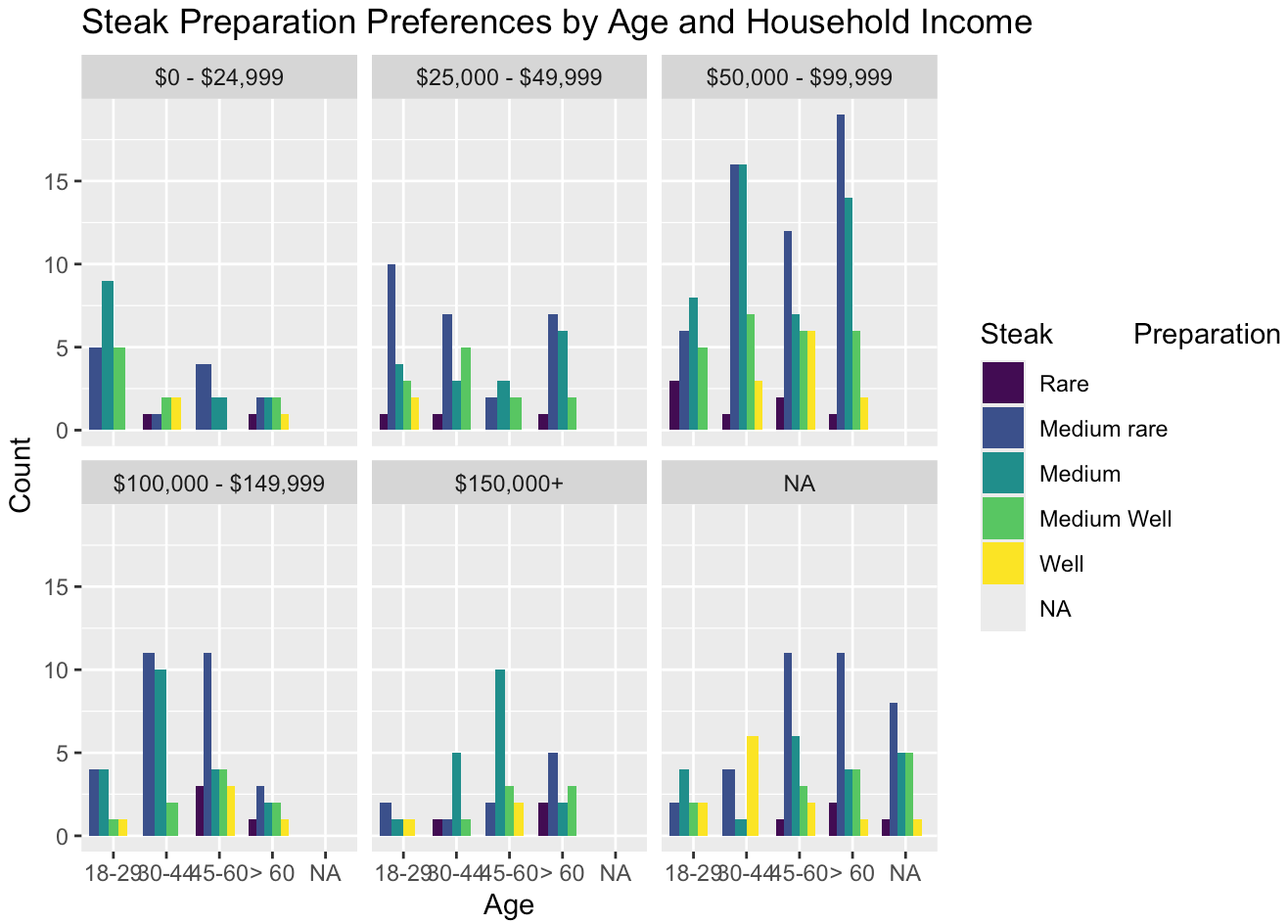
a.) age

```
library(ggplot2)
ggplot(steak_survey, aes(x = age, fill = steak_prep)) + geom_bar(position = "dodge") +
  labs(title = "Steak Preparation Preferences by Age", x = "Age", y = "Count", fill = "S
teak Preparation")
```



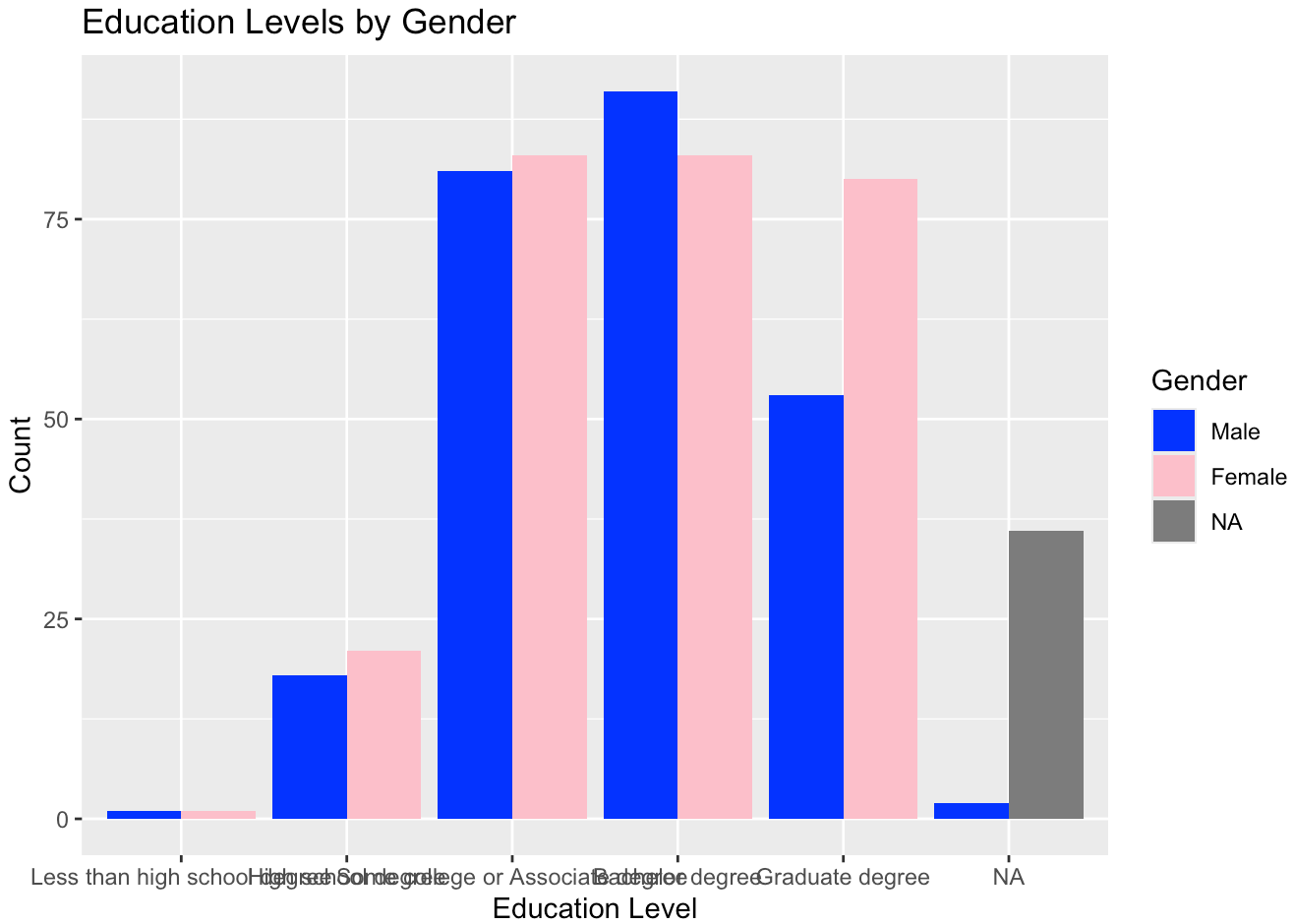

b.) age and hhold income

```
ggplot(steak_survey, aes(x = age, fill = steak_prep)) + geom_bar(position = "dodge") + f
acet_wrap(~ hhold_income) +
  labs(title = "Steak Preparation Preferences by Age and Household Income", x = "Age", y
= "Count", fill = "Steak          Preparation")
```




c.) gender and education

```
ggplot(steak_survey, aes(x = educ, fill = as.factor(female))) + geom_bar(position = "dod
ge") +
  labs(title = "Education Levels by Gender", x = "Education Level", y = "Count",fill =
"Gender") +
  scale_fill_manual(values = c("FALSE" = "blue", "TRUE" = "pink"),
                    labels = c("Male", "Female"))
```




# Bad Drivers,Problem 3

Driving behavior patterns

```
head(bad_drivers)
```

```
## # A tibble: 6 × 8
##   state      num_drivers perc_speeding perc_alcohol perc_not_distracted
##   <chr>            <dbl>         <int>        <int>               <int>
## 1 Alabama           18.8            39           30                  96
## 2 Alaska            18.1            41           25                  90
## 3 Arizona           18.6            35           28                  84
## 4 Arkansas          22.4            18           26                  94
## 5 California        12              35           28                  91
## 6 Colorado          13.6            37           28                  79
## # ℹ 3 more variables: perc_no_previous <int>, insurance_premiums <dbl>,
## #   losses <dbl>
```

```
help("bad_drivers")
```

a.) Which state had the highest number of accidents per billion miles? Which had the lowest?

```
highest_accident_state <- bad_drivers |>
  filter(num_drivers == max(num_drivers)) |>
  select(state, num_drivers)

lowest_accident_state <- bad_drivers |>
  filter(num_drivers == min(num_drivers)) |>
  select(state, num_drivers)

print(highest_accident_state)
```

```
## # A tibble: 2 × 2
##   state          num_drivers
##   <chr>                <dbl>
## 1 North Dakota          23.9
## 2 South Carolina        23.9
```

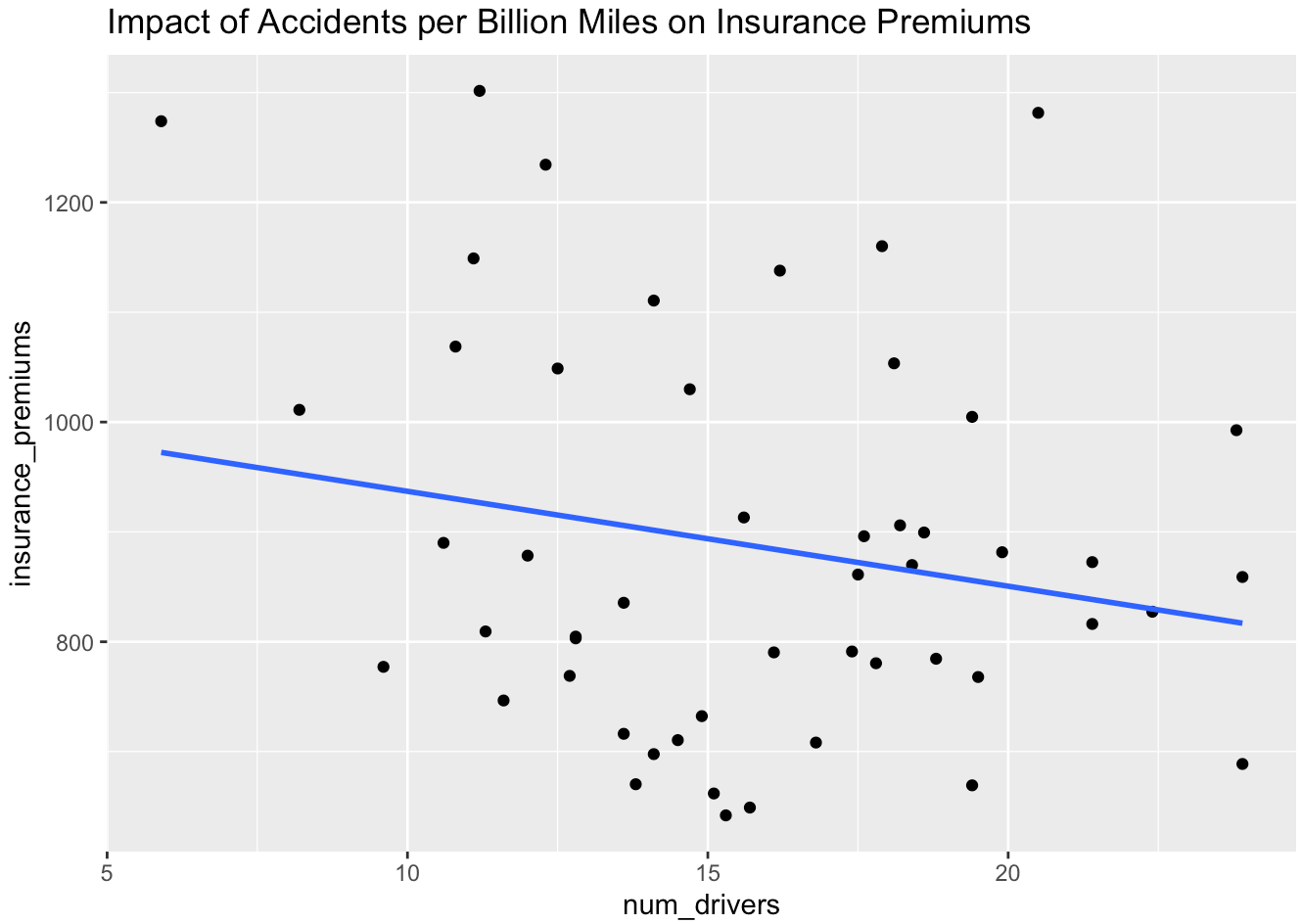
```
print(lowest_accident_state)
```

```
## # A tibble: 1 × 2
##   state                num_drivers
##   <chr>                      <dbl>
## 1 District of Columbia         5.9
```

b.) Create a graphic that shows the impact of accidents per billion miles on insurance premiums?

```
ggplot(bad_drivers, aes(x = num_drivers, y = insurance_premiums)) +
  geom_point() + geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Impact of Accidents per Billion Miles on Insurance Premiums", x = "num_d
rivers", y = "insurance_premiums")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

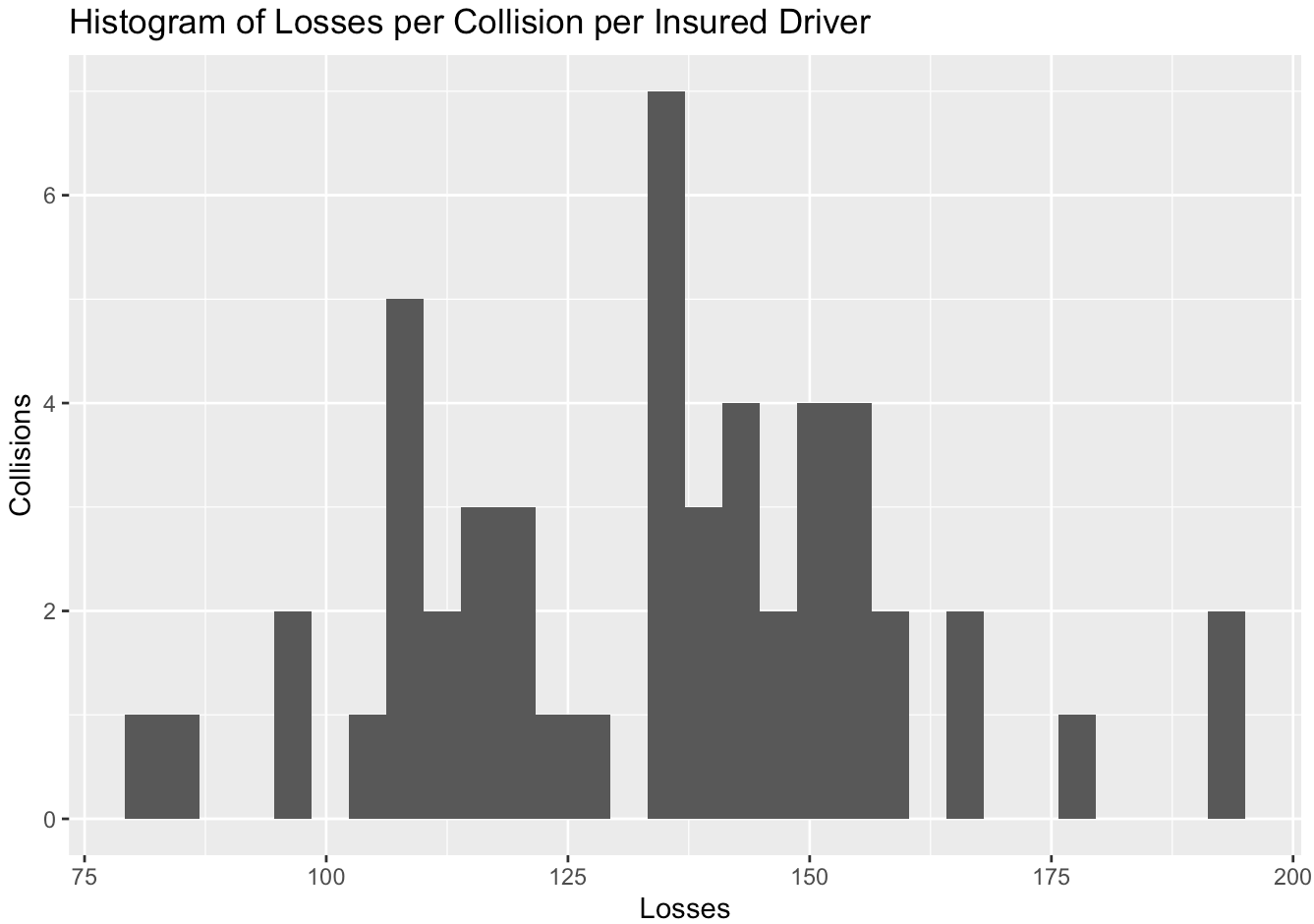



c.)

Create a histogram of losses per collision per insured driver

```
ggplot(bad_drivers, aes(x = losses)) +
  geom_histogram() +
  labs(title = "Histogram of Losses per Collision per Insured Driver", x = "Losses", y =
"Collisions")
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




d.) Create a graphic that shows if alcohol use in fatal collisions is related to speeding in fatal collisions

```
ggplot(bad_drivers, aes(x = perc_alcohol, y = perc_speeding)) +
  geom_point() + geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Relationship Between Alcohol Use and Speeding in Fatal Collisions", x =
"perc_alcohol", y = "perc_speeding")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

# Palmer Penguins Data Set Problem 4

```
library("palmerpenguins")
library(dplyr)
library(tidyr)
```

```
head(penguins)
```

```
## # A tibble: 6 × 8
##   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>              <dbl>         <dbl>             <int>       <int>
## 1 Adelie  Torgersen           39.1          18.7               181        3750
## 2 Adelie  Torgersen           39.5          17.4               186        3800
## 3 Adelie  Torgersen           40.3          18                 195        3250
## 4 Adelie  Torgersen           NA            NA                  NA          NA
## 5 Adelie  Torgersen           36.7          19.3               193        3450
## 6 Adelie  Torgersen           39.3          20.6               190        3650
## # ℹ 2 more variables: sex <fct>, year <int>
```

```
help(penguins)
```

a.) Create a long version of the data set, with species, island and sex as indices (variables), the bill length, bill depth, flipper length and body mass should be in the names_to setting and the values in values, year should be dropped

It should look like this

Species island sex Variable Value Adelie Torgerson male bill_lenth_mm 39.1 Adelie Torgerson male bill_depth_mm 18.7 Adelie Torgerson male flipper_length_mm 181 Adelie Torgerson male body_mass_g 3750 Adelie Torgerson female bill_lenth_mm 39.5 Adelie Torgerson female bill_depth_mm 17.4 Adelie Torgerson female flipper_length_mm 18 Adelie Torgerson female body_mass_g 3800

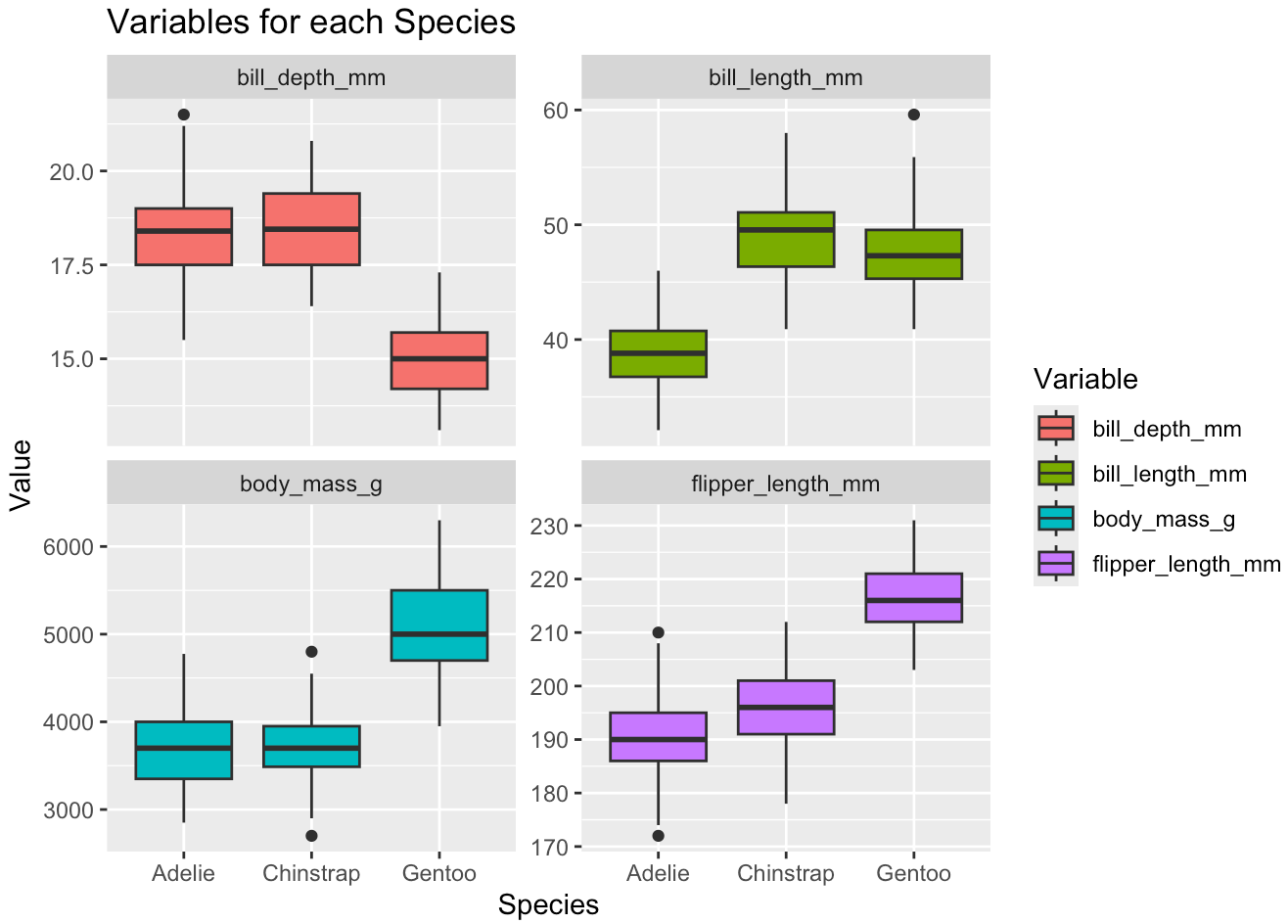
```
penguins_long <- penguins %>%
  select(-year) %>%
  pivot_longer(
    cols = c(bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g),
    names_to = "Variable",
    values_to = "Value"
  )

penguins_long
```

```
## # A tibble: 1,376 × 5
##    species island    sex    Variable           Value
##    <fct>   <fct>     <fct>  <chr>              <dbl>
##  1 Adelie  Torgersen male   bill_length_mm      39.1
##  2 Adelie  Torgersen male   bill_depth_mm       18.7
##  3 Adelie  Torgersen male   flipper_length_mm  181
##  4 Adelie  Torgersen male   body_mass_g       3750
##  5 Adelie  Torgersen female bill_length_mm      39.5
##  6 Adelie  Torgersen female bill_depth_mm       17.4
##  7 Adelie  Torgersen female flipper_length_mm  186
##  8 Adelie  Torgersen female body_mass_g       3800
##  9 Adelie  Torgersen female bill_length_mm      40.3
## 10 Adelie  Torgersen female bill_depth_mm       18
## # ℹ 1,366 more rows
```

b.) Produce a box plot that shows the all the Variables (bill_length, bill_depth,flipper_length and body_mass) for each species, using the long data

```
ggplot(penguins_long, aes(x = species, y = Value, fill = Variable)) + geom_boxplot() +
  facet_wrap(~ Variable, scales = "free_y") +
  labs(title = "Variables for each Species", x = "Species", y = "Value", fill = "Variabl
e")
```

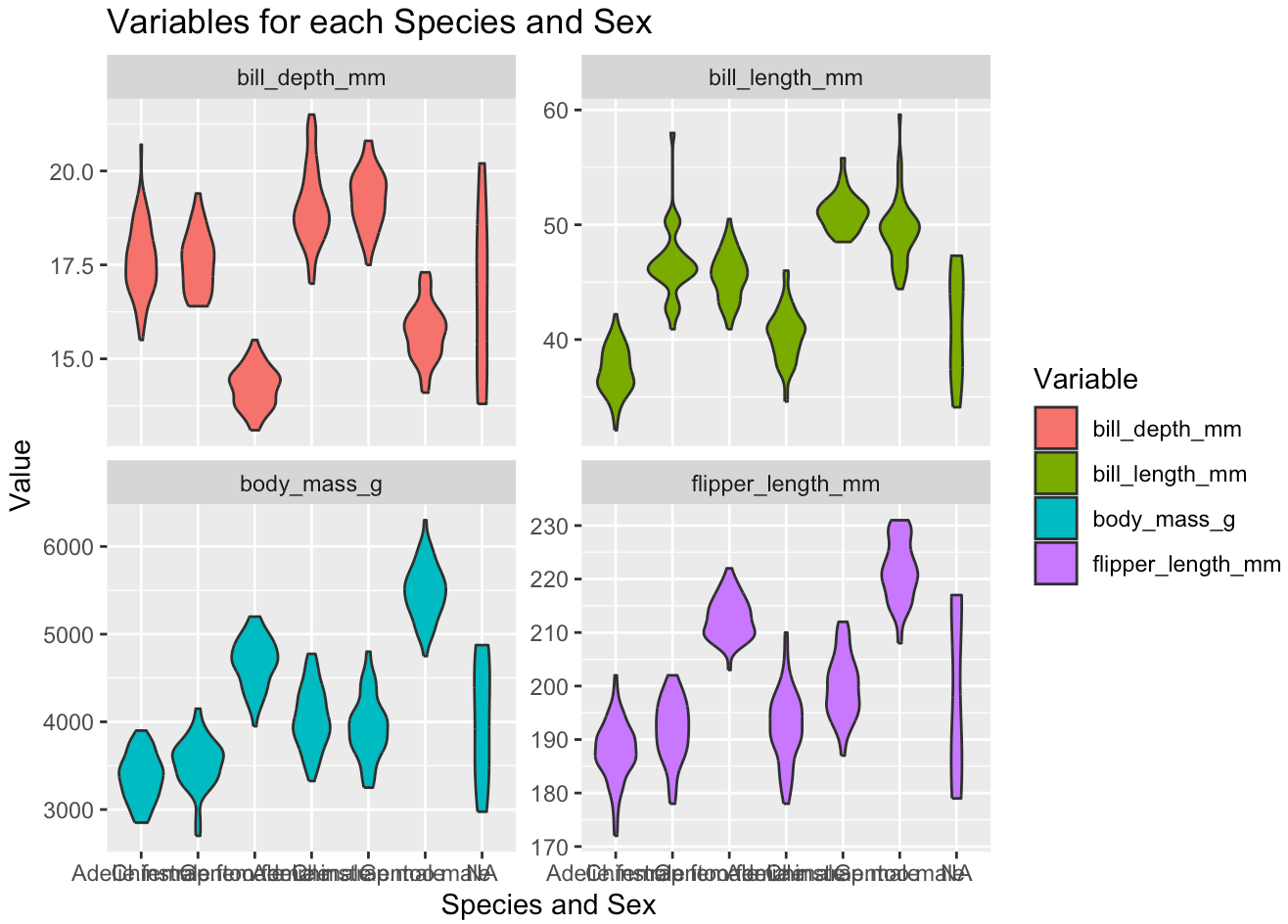
```
## Warning: Removed 8 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




c.) Show a violin plot that shows all the variables (bill_length, bill_depth,flipper_length and body_mass) for each species and sex, using the long data

```
ggplot(penguins_long, aes(x = interaction(species, sex), y = Value, fill = Variable)) +
geom_violin() +
  facet_wrap(~ Variable, scales = "free_y") +
   labs(title = "Variables for each Species and Sex", x = "Species and Sex", y = "Valu
e", fill = "Variable")
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## (`stat_ydensity()`).
```




# Palmer Penguins Data Set Problem 5

```
library("palmerpenguins")
```

```
head(penguins)
```

```
## # A tibble: 6 × 8
##   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>              <dbl>         <dbl>             <int>       <int>
## 1 Adelie  Torgersen           39.1          18.7               181        3750
## 2 Adelie  Torgersen           39.5          17.4               186        3800
## 3 Adelie  Torgersen           40.3          18                 195        3250
## 4 Adelie  Torgersen           NA            NA                  NA          NA
## 5 Adelie  Torgersen           36.7          19.3               193        3450
## 6 Adelie  Torgersen           39.3          20.6               190        3650
## # ℹ 2 more variables: sex <fct>, year <int>
```

```
help("penguins")
```

a.) How many species and islands are there (show code that calculates this, don't count from a table)

```
species <- penguins %>%
  summarise(unique_species = n_distinct(species))

islands <- penguins %>%
  summarise(unique_islands = n_distinct(island))

species
```

```
## # A tibble: 1 × 1
##   unique_species
##            <int>
## 1              3
```

```
islands
```

```
## # A tibble: 1 × 1
##   unique_islands
##            <int>
## 1              3
```

b.) Create at table that shows for each species, the mean body mass by sex

```
penguins |> group_by(species, sex) |>summarize (mean_body_mass=mean(body_mass_g, na.rm =
TRUE))
```

```
## `summarise()` has grouped output by 'species'. You can override using the
## `.groups` argument.
```

```
## # A tibble: 8 × 3
## # Groups:   species [3]
##   species   sex    mean_body_mass
##   <fct>     <fct>           <dbl>
## 1 Adelie    female          3369.
## 2 Adelie    male            4043.
## 3 Adelie    <NA>            3540
## 4 Chinstrap female          3527.
## 5 Chinstrap male            3939.
## 6 Gentoo    female          4680.
## 7 Gentoo    male            5485.
## 8 Gentoo    <NA>            4588.
```

c.) For the heaviest penguin species, make a table that shows body mass by island and year

```
heaviest_species <- penguins |> group_by(species) |> summarize(mean_body_mass = mean(bod
y_mass_g, na.rm = TRUE)) |>
  arrange(desc(mean_body_mass)) |> slice(1) |> pull(species)

body_mass_table <- penguins |>
  filter(species == heaviest_species) |>
  group_by(island, year) |>
  summarize(mean_body_mass = mean(body_mass_g, na.rm = TRUE), .groups = "drop")

body_mass_table
```

```
## # A tibble: 3 × 3
##   island  year mean_body_mass
##   <fct>  <int>          <dbl>
## 1 Biscoe  2007          5071.
## 2 Biscoe  2008          5020.
## 3 Biscoe  2009          5141.
```

# Exponential distribution Problem 6

We will be looking at the exponential distribution, which is a continuous distribution much like the normal or poison

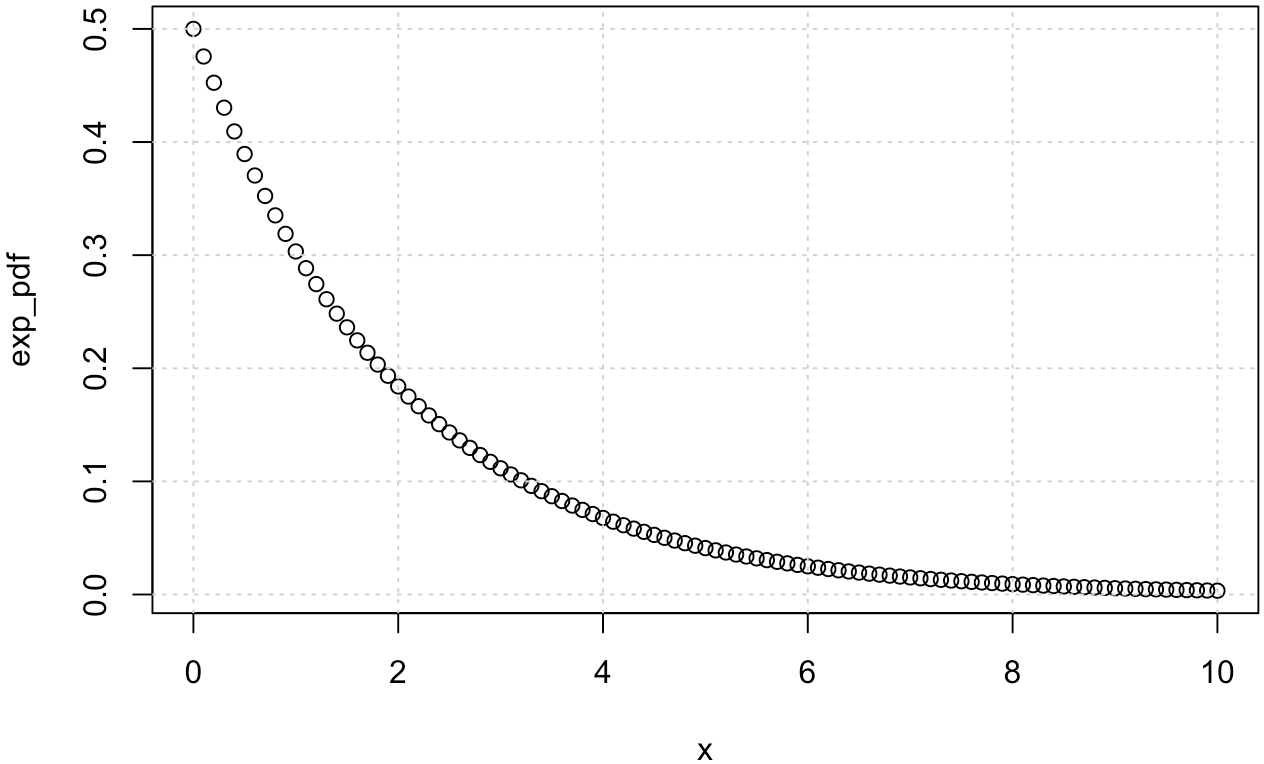
It has only one parameter a rate.

We will look at an exponential distribution with a rate of .5

Using a sequence of x values form 0 to 10 in steps of 0.1 (set up in code block below)

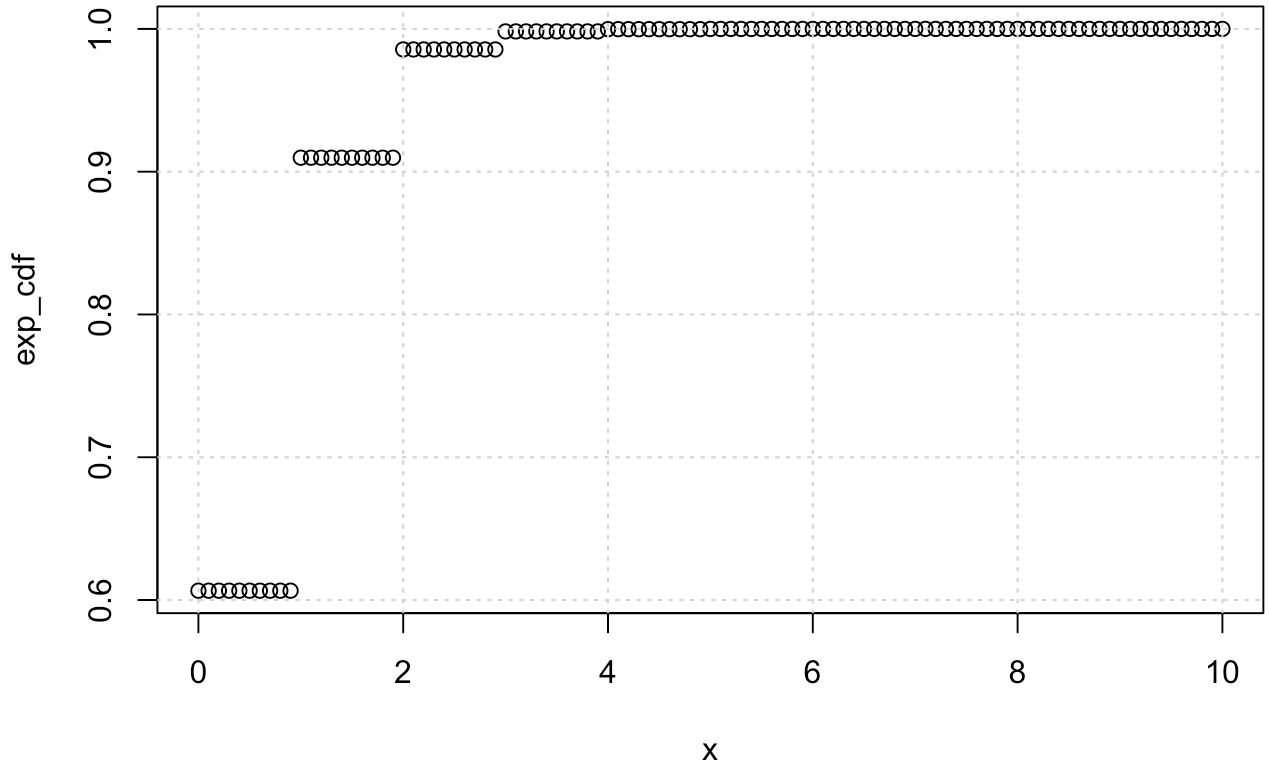
a.) Plot the probability distribution function (pdf) for an exponential distribution with rate 0.5

```
x=seq(0,10,0.1)

lambda= 0.5

exp_pdf=dexp(x, lambda)

plot(x,exp_pdf)
grid()
```




b.) Plot the cummulative probability distribution (cdf) for an exponential distribution with rate 0.5, add a grid to this

```
x=seq(0,10,0.1)

lambda= 0.5

exp_cdf=ppois(x,lambda)

plot(x,exp_cdf)

grid()
```




c.) Find the mean and standard deviation of this distribution #The mean and standard deviation are both 2.

```
1/0.5
```

```
## [1] 2
```

d.) What is the probablity that a value (x) from this distribution is greater than 2 #About 36.79%

```
#greater than is 1- and less than is plain

x <- 2

1 - pexp(x, rate = 0.5)
```

```
## [1] 0.3678794
```

e.) What is the probability that value (x) from this distribution is in the range 1 to 3?

```
lambda= 0.5

pexp(3, 0.5) - pexp(1, 0.5)
```

```
## [1] 0.3834005
```

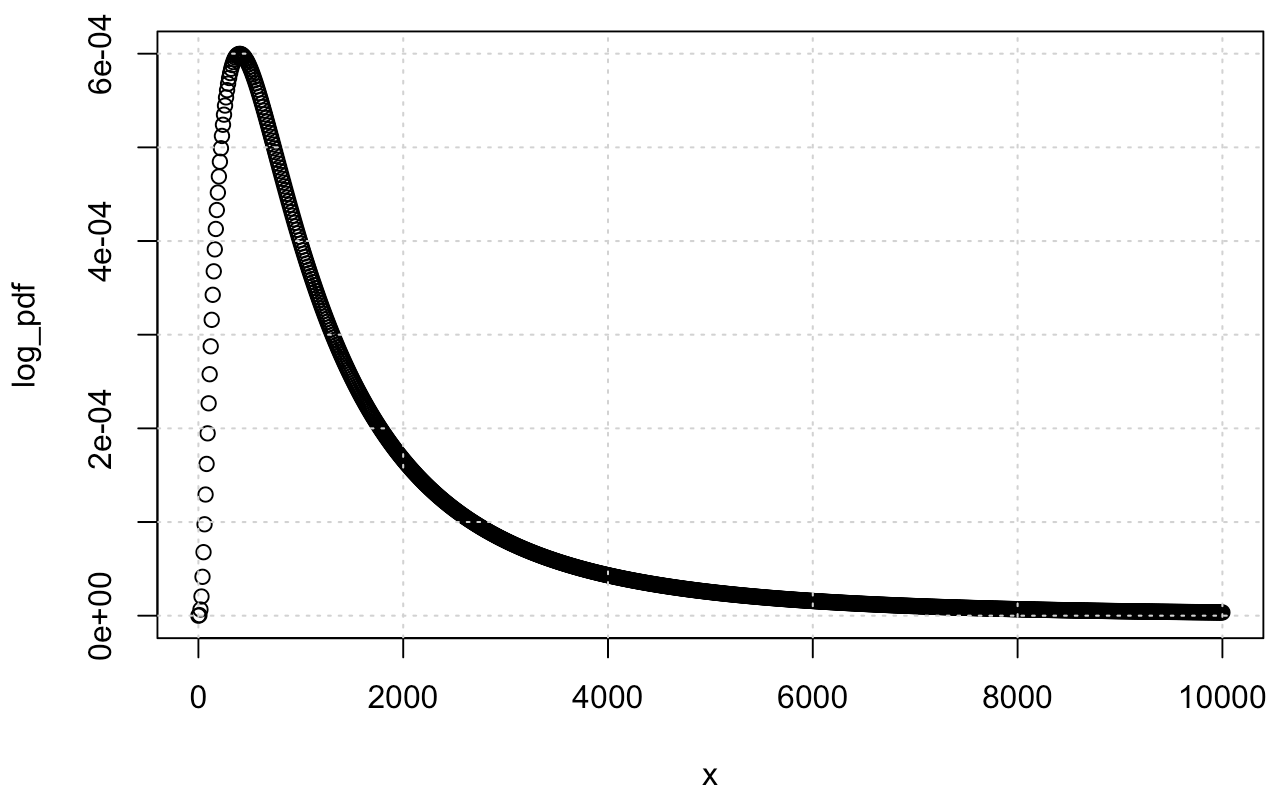
# Log normal Distribution Problem #7

The log normal has parameters meanlog and sdlog

We will set meanlog=7, sdlog=1

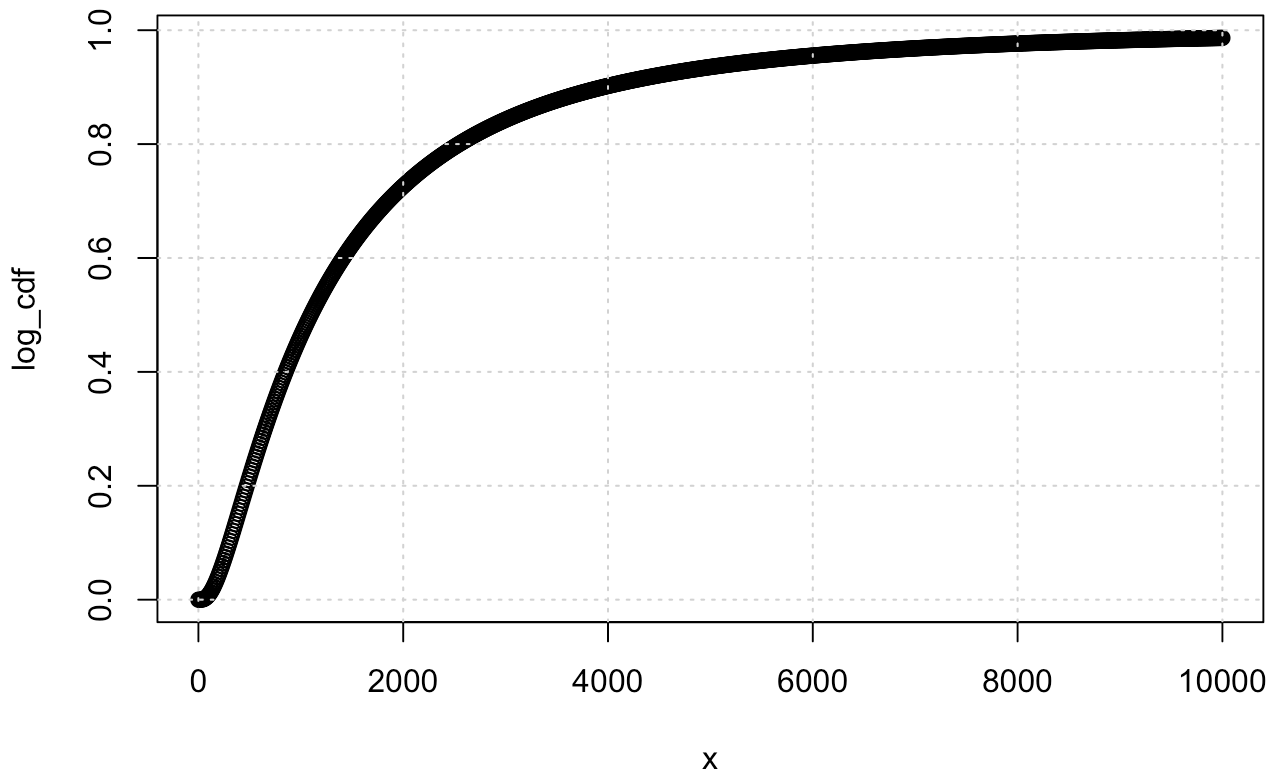
a.) Plot the log normal probability distribution function (pdf) for x=0 to 10,000 using the x sequence list set up below

```
x=seq(0,10000,10)

meanlog <- 7
sdlog <- 1

log_pdf=dlnorm(x, meanlog = meanlog, sdlog = sdlog)

plot(x,log_pdf)

grid()
```




b.) Plot the log normal cummulative distributin function (cdf) for the same range of x. Add a grid to this plot

```
x=seq(0,10000,10)

meanlog <- 7
sdlog <- 1

log_cdf=plnorm(x, meanlog = meanlog, sdlog = sdlog)

plot(x,log_cdf)

grid()
```




c.) What are the mean and standard deviation of this distribution

```
meanlog <- 7
sdlog <- 1

mean_value <- exp(meanlog + 0.5*sdlog^2)

sd_value <- sqrt((exp(sdlog^2) - 1) * exp(2*meanlog + sdlog^2))

mean_value
```

```
## [1] 1808.042
```

```
sd_value
```

```
## [1] 2370.041
```

d.) What is the maximum of the pdf?

```
x=seq(0,10000,10)

meanlog <- 7
sdlog <- 1

log_pdf=dlnorm(x, meanlog = meanlog, sdlog = sdlog)

max_pdf <- max(log_pdf)

max_pdf
```

```
## [1] 0.0005997636
```

e.) What is the probability of a random x from this distribution being less than 500? #About 21.61%

```
#greater than is 1- and less than is plain

meanlog <- 7
sdlog <- 1

plnorm(500, meanlog = meanlog, sdlog = sdlog)
```

```
## [1] 0.2161119
```

f.) What is the probability of a random x from this distribution being less than 200 OR more than 1500? #About 42.15%

```
meanlog <- 7
sdlog <- 1

plnorm(200, meanlog = meanlog, sdlog = sdlog) + 1 - plnorm(1500, meanlog = meanlog, sdlo
g = sdlog)
```

```
## [1] 0.4214641
```

#Midterm Quiz ?'s

# Palmer Penguins Data Set Problem 5

We will change and use the larger Hawks data set

```
#install.packages("Stat2Data")
library("Stat2Data")
data(Hawks)
```

```
head(Hawks)
```

```
##   Month Day Year CaptureTime ReleaseTime  BandNumber Species Age Sex Wing
## 1     9  19 1992       13:30               877-76317      RT   I      385
## 2     9  22 1992       10:30               877-76318      RT   I      376
## 3     9  23 1992       12:45               877-76319      RT   I      381
## 4     9  23 1992       10:50               745-49508      CH   I   F  265
## 5     9  27 1992       11:15              1253-98801      SS   I   F  205
## 6     9  28 1992       11:25             1207-55910       RT   I      412
##   Weight Culmen Hallux Tail StandardTail Tarsus WingPitFat KeelFat Crop
## 1    920   25.7   30.1  219           NA     NA         NA      NA   NA
## 2    930     NA     NA  221           NA     NA         NA      NA   NA
## 3    990   26.7   31.3  235           NA     NA         NA      NA   NA
## 4    470   18.7   23.5  220           NA     NA         NA      NA   NA
## 5    170   12.5   14.3  157           NA     NA         NA      NA   NA
## 6   1090   28.5   32.2  230           NA     NA         NA      NA   NA
```

You may need to use na.rm=TRUE with this data set

a.) How many species and age categories are there (show code that calculates this, don't count from a table)

```
Species_hawks <- Hawks %>%
  summarise(unique_Species = n_distinct(Species))

Age_hawks <- Hawks %>%
  summarise(unique_Age = n_distinct(Age))

Species_hawks
```

```
##   unique_Species
## 1              3
```

```
Age_hawks
```

```
##   unique_Age
## 1          2
```

b.) Create at table that shows for each species, the mean Weight by sex

```
Hawks |> group_by(Species, Sex) |>summarize (mean_Weight=mean(Weight, na.rm = TRUE))
```

```
## `summarise()` has grouped output by 'Species'. You can override using the
## `.groups` argument.
```

```
## # A tibble: 9 × 3
## # Groups:   Species [3]
##   Species Sex   mean_Weight
##   <fct>   <fct>       <dbl>
## 1 CH      ""           402
## 2 CH      "F"          490.
## 3 CH      "M"          348.
## 4 RT      ""          1094.
## 5 RT      "F"         1147.
## 6 RT      "M"         1080
## 7 SS      ""            95
## 8 SS      "F"          175.
## 9 SS      "M"          119.
```

c.) For the heaviest hawk species, make a table that shows weight by Sex and year

```
heaviest_hawk_species <- Hawks |> group_by(Species) |> summarize(mean_Weight = mean(Weig
ht, na.rm = TRUE)) |>
  arrange(desc(mean_Weight)) |> slice(1) |> pull(Species)

Weight_table <- Hawks |>
  filter(Species == heaviest_hawk_species) |>
  group_by(Sex, Year) |>
  summarize(mean_Weight = mean(Weight, na.rm = TRUE), .groups = "drop")

Weight_table
```

```
## # A tibble: 14 × 3
##    Sex    Year mean_Weight
##    <fct> <int>       <dbl>
##  1 ""     1992       1098.
##  2 ""     1993       1172.
##  3 ""     1994       1099.
##  4 ""     1995       1087.
##  5 ""     1996       1115.
##  6 ""     1997       1038
##  7 ""     1998       1189
##  8 ""     1999       1085.
##  9 ""     2000       1071.
## 10 ""     2001       1114.
## 11 ""     2002       1081.
## 12 ""     2003       1088.
## 13 "F"    2000       1147.
## 14 "M"    1999       1080
```

#Driving behavior patterns

```
library("fivethirtyeight")
```

```
head(bad_drivers)
```

```
## # A tibble: 6 × 8
##   state      num_drivers perc_speeding perc_alcohol perc_not_distracted
##   <chr>            <dbl>         <int>        <int>               <int>
## 1 Alabama           18.8            39           30                  96
## 2 Alaska            18.1            41           25                  90
## 3 Arizona           18.6            35           28                  84
## 4 Arkansas          22.4            18           26                  94
## 5 California        12              35           28                  91
## 6 Colorado          13.6            37           28                  79
## # ℹ 3 more variables: perc_no_previous <int>, insurance_premiums <dbl>,
## #   losses <dbl>
```

```
help(bad_drivers)
```

a.) Which state had the highest percent speeding? Which had the lowest?

```
highest_speeding_state <- bad_drivers |>
  filter(perc_speeding == max(perc_speeding)) |>
  select(state, perc_speeding)

lowest_speeding_state <- bad_drivers |>
  filter(perc_speeding == min(perc_speeding)) |>
  select(state, perc_speeding)

print(highest_speeding_state)
```

```
## # A tibble: 1 × 2
##   state  perc_speeding
##   <chr>          <int>
## 1 Hawaii            54
```

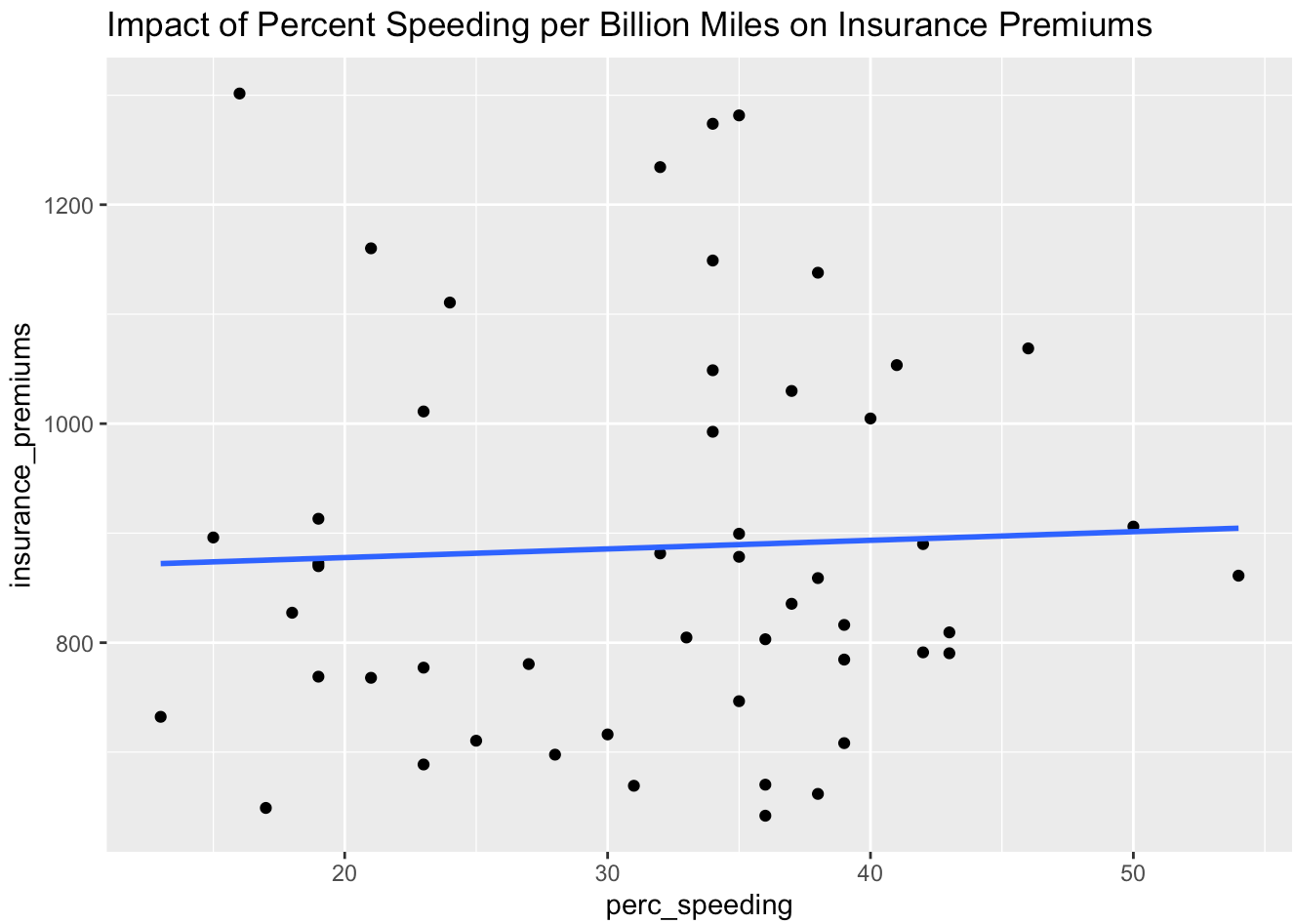
```
print(lowest_speeding_state)
```

```
## # A tibble: 1 × 2
##   state    perc_speeding
##   <chr>            <int>
## 1 Nebraska            13
```

b.) Create a graphic that shows the impact of percent speeding on insurance premiums?

```
ggplot(bad_drivers, aes(x = perc_speeding, y = insurance_premiums)) +
  geom_point() + geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Impact of Percent Speeding per Billion Miles on Insurance Premiums", x =
"perc_speeding", y = "insurance_premiums")
```

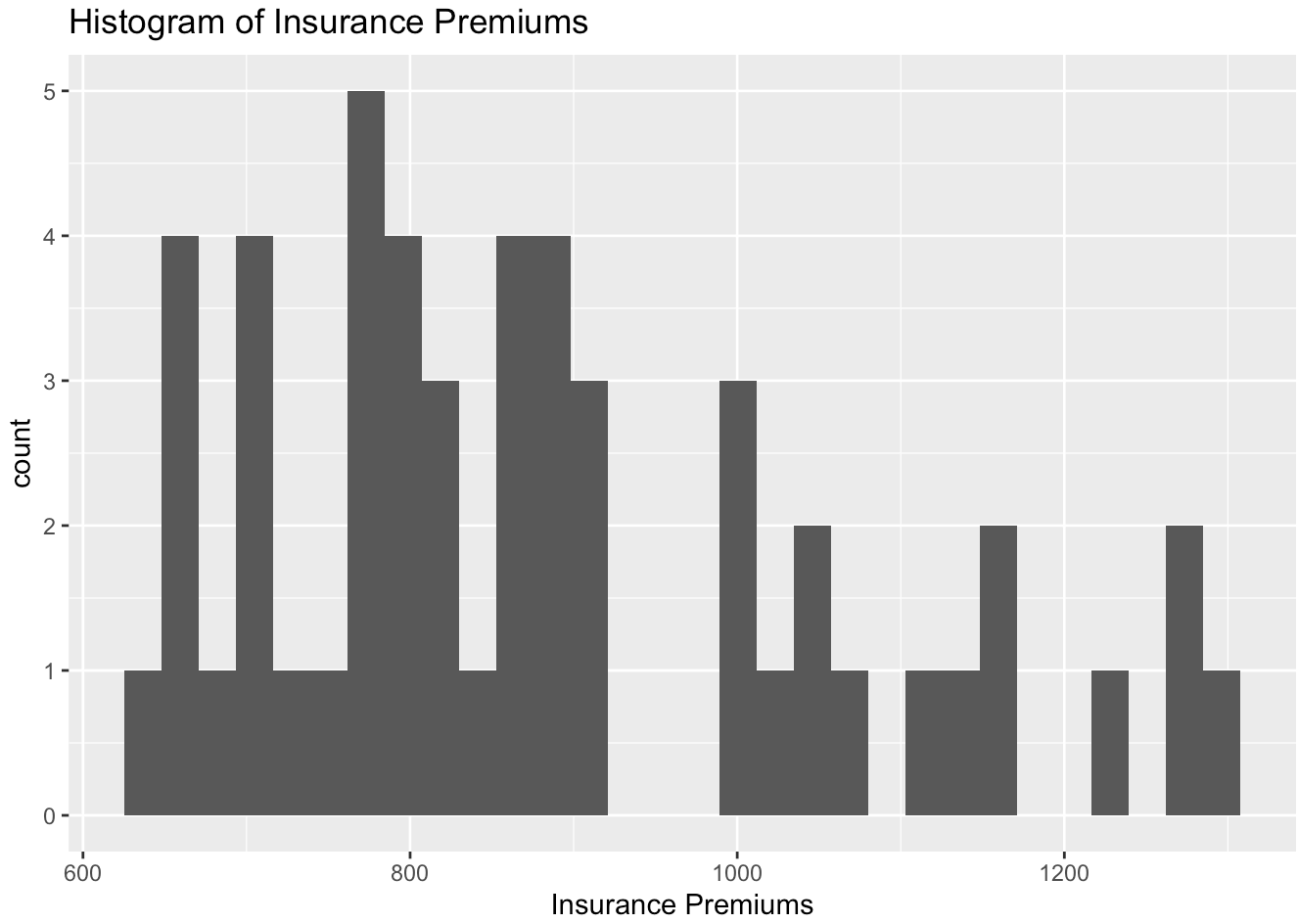
```
## `geom_smooth()` using formula = 'y ~ x'
```




c.) Create a histogram of insurance premiums

```
ggplot(bad_drivers, aes(x = insurance_premiums)) +
  geom_histogram() +
  labs(title = "Histogram of Insurance Premiums", x = "Insurance Premiums")
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




d.) Create a graphic that shows if alcohol use in fatal collisions is related to speeding in fatal collisions

```
ggplot(bad_drivers, aes(x = perc_alcohol, y = perc_speeding)) +
  geom_point() + geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Relationship Between Alcohol Use and Speeding in Fatal Collisions", x =
"perc_alcohol", y = "perc_speeding")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

# Exponential distribution Problem 6

We will be looking at the exponential distribution, which is a continuous distribution much like the normal or poison

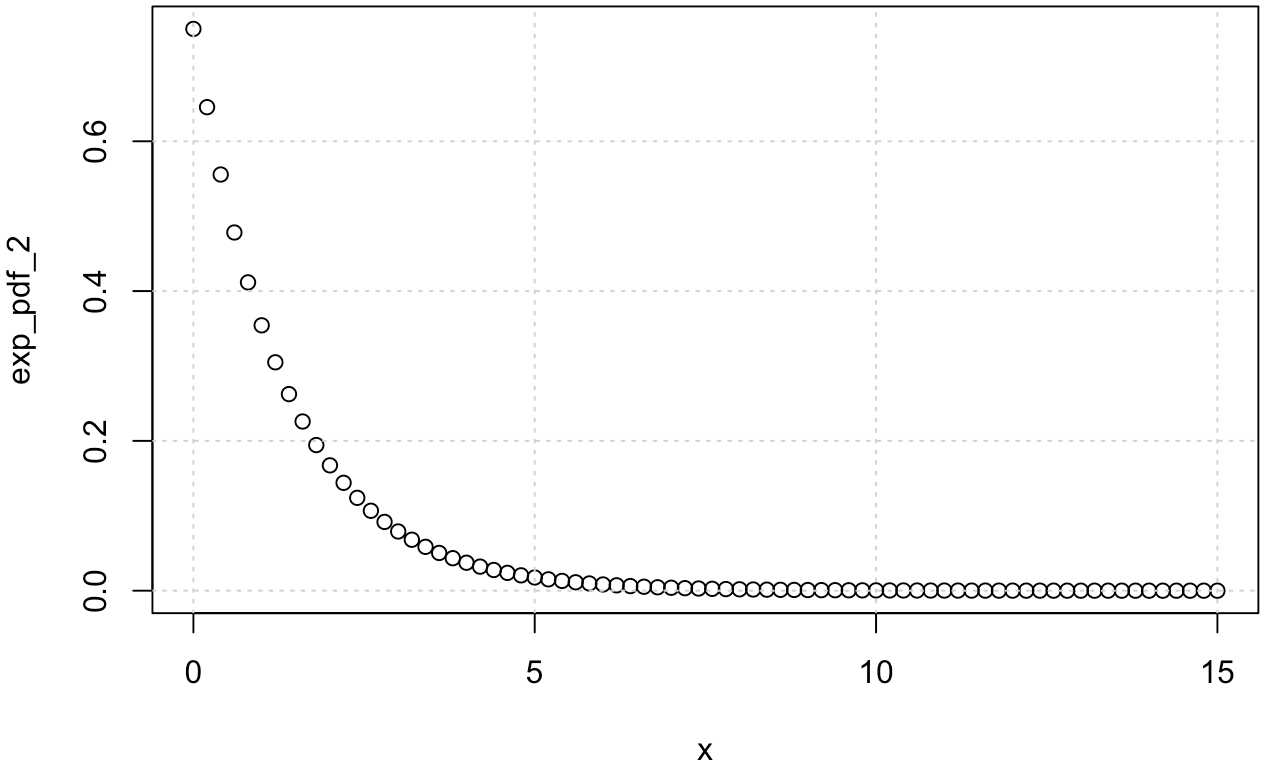
It has only one parameter a rate.

We will look at an exponential distribution with a rate of .75

Using a sequence of x values form 0 to 15 in steps of 0.2 (set up in code block below)

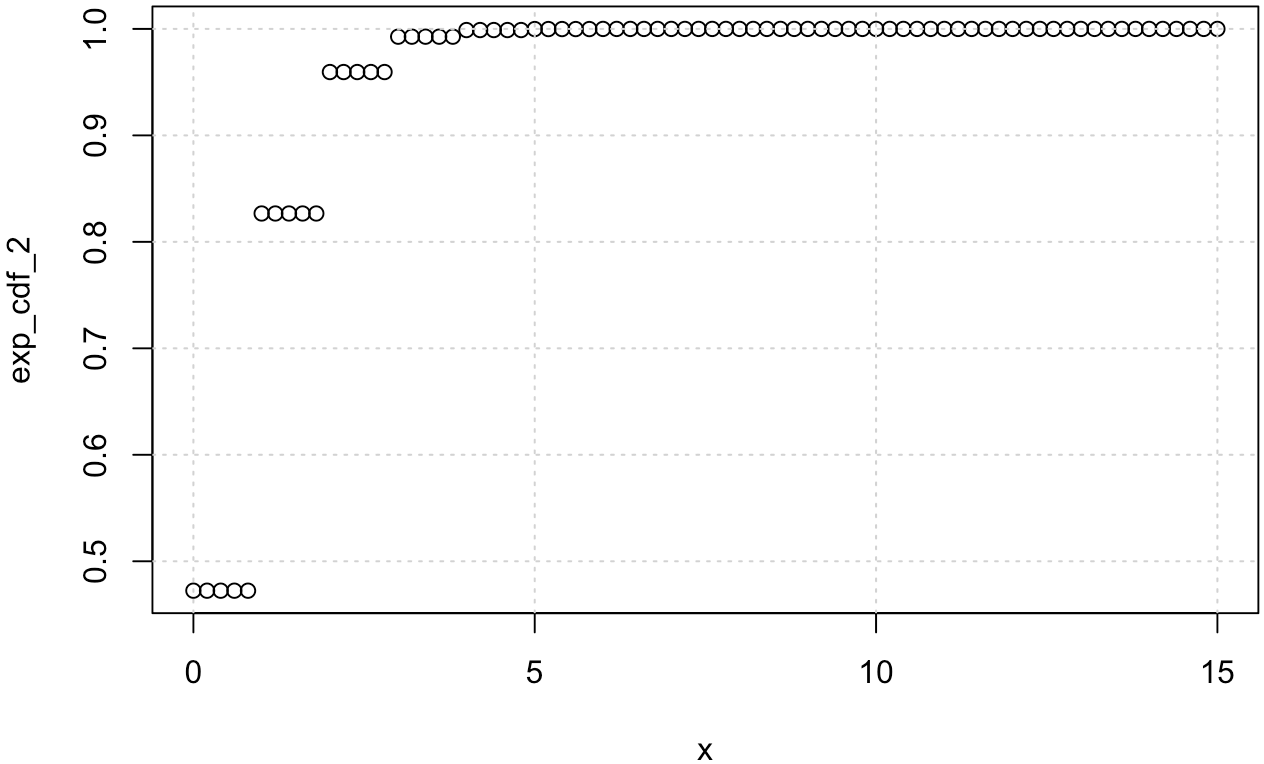
a.) Plot the probability distribution function (pdf) for an exponential distribution with rate 0.75

```
x=seq(0,15,0.2)

lambda= 0.75

exp_pdf_2=dexp(x, lambda)

plot(x,exp_pdf_2)
grid()
```




b.) Plot the cummulative probability distribution (cdf) for an exponential distribution with rate 0.75, add a grid to this

```
x=seq(0,15,0.2)

lambda= 0.75

exp_cdf_2=ppois(x,lambda)

plot(x,exp_cdf_2)

grid()
```




c.) Find the mean and standard deviation of this distribution #The mean and standard deviation are both 1.333333.

```
1/0.75
```

```
## [1] 1.333333
```

d.) What are the 25% and 75% quantiles values? #The 25% quartile is 0.3835761 and the 75% quartile is 1.848392.

```
quartile_25p <- qexp(0.25, rate = 0.75)

quartile_75p <- qexp(0.75, rate = 0.75)

quartile_25p
```

```
## [1] 0.3835761
```

```
quartile_75p
```

```
## [1] 1.848392
```

e.) What is the probablity that a value (x) from this distribution is less than 1.8 #About 74.08%

```
x <- 1.8

pexp(x, rate = 0.75)
```

```
## [1] 0.7407597
```

f.) What is the probability that value (x) from this distribution is in the range 2 to 4? #About 17.33%.

```
lambda= 0.75

pexp(4, 0.75) - pexp(2, 0.75)
```

```
## [1] 0.1733431
```